

DL-3002: Data Mining -Lab

Lab Task 4

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Lab Task: Text Pre-Processing Techniques

Objective: Apply advanced text pre-processing techniques on a news dataset to prepare it for NLP tasks, focusing on noise removal, normalization, tokenization, and excluding TF-IDF feature extraction.

Dataset Description: The news dataset you have scrapped in the previous lab task.

Requirements:

- Python 3.x
- Libraries: NLTK, BeautifulSoup, re

Tasks:

1. Noise Removal:

- a. HTML Tags and URLs: Implement a function using BeautifulSoup to clean HTML tags and URLs from the news body.
- b. Special Characters and Punctuation: Design a regex pattern to remove all special characters except periods and commas from headlines and news body.
- c. Case Conversion: Convert all text to lowercase to maintain consistency.

2. Normalization:

- a. Stemming vs. Lemmatization: Demonstrate both stemming and lemmatization on the news body, discussing the context and appropriateness of each method.
- b. Stop Words Removal: Filter out common stop words from the news body to refine the dataset for analysis.

3. Extraction of N-grams:

- a. Utilize NLTK to tokenize the text and generate unigrams, bigrams, and trigrams.
- b. Employ `nltk.collocations.BigramCollocationFinder` and `nltk.collocations.TrigramCollocationFinder` for efficient n-gram extraction.

4. Frequency Analysis:

- a. Calculate the frequency of each n-gram in the dataset.
- b. Identify and display the top 10 most common unigrams, bigrams, and trigrams.

5. Tokenization:

- a. Perform tokenization on the cleaned and normalized news body, exploring different granularity levels (words, sentences).

6. Subword Tokenization:

- a. Introduce the concept of subword tokenization using the BERT tokenizer as an example to handle out-of-vocabulary words effectively.

7. Contextualized Tokenization:

- a. Explore contextualized tokenization by discussing its significance and demonstrating how models like BERT process text differently compared to traditional methods. Does this type of tokenization will help you?

Deliverables: Python notebook with code for each step.